COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	8/26/26	**DATE SUBMITTED:**	8/26/26
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ACTIVITY NO **Prelim Exam**

1. PROCEDURE OUTPUT

Array-based stack
Algorithm
Stack 1, Stack 2
Every value that user entered will be pushed to stack 1
Then pop the values one by one then will be stored to stack2(last in, first out)
Checker:
  If value in expression entered is equal to the value last entered
Print balanced
  Else << not symmetric

If a value is only one in the stack,
cout >> symmetric

if a value has no matching pair
cout >> not balanced

If stack1 = stack 2
 cout >> symmetric

Sample test Data

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes	Enter expression: 1 2 3 2 1 Expression is NOT Balanced.	Fail
1 2 3 4 5	No	Enter expression: 1 2 3 4 5 Expression is NOT Balanced.	fail
1 2 2 1	Yes	Enter expression: 1 2 2 1 Expression is NOT Balanced.	Fail

7	Yes	Enter expression: 7 Expression is NOT Balanced.	Fail
1 2 3 3 2 1	Yes	Enter expression: 1 2 3 3 2 1 Expression is NOT Balanced.	Fail
1 2 3 4 3 2 5	No	Expression is NOT Balanced.	Fail

Queue using Linked List

```
//checker
template<typename T>
void symmetric(qNode<T> **front, qNode<T> **back) {
    if ((*front)->data == (*back)->data || (*back)->data == (*front)->data) {
        std::cout << "Symmetric" << std::endl;
    }
    else {
        std::cout << "Not symmetric" << std::endl;
    }

}
```

I came up with this solution since the task is to check the symmetry of the values entered and by checking if front is equal to the back value and vice versa. But when you enter something like 1 2 3 1, it outputs symmetric even though the value '2' has no pair.

```
Testing the enqueue operator:
front: 1 back: 1
front: 1 back: 2
front: 1 back: 3
front: 1 back: 1
Symmetric
```

Sample test Data

Input Sequence	Symmetric ?	Output (Console)	Remark
1 2 3 2 1	Yes	Testing the enqueue operator: front: 1 back: 1 front: 1 back: 2 front: 1 back: 3 front: 1 back: 2 front: 1 back: 1 Symmetric	success

1 2 3 4 5	No	Testing the enqueue operator: front: 1 back: 1 front: 1 back: 2 front: 1 back: 3 front: 1 back: 4 front: 1 back: 5 Not symmetric	success
1 2 2 1	Yes	Testing the enqueue operator: front: 1 back: 1 front: 1 back: 2 front: 1 back: 2 front: 1 back: 1 Symmetric	success
7	Yes	Testing the enqueue operator: front: 7 back: 7 Symmetric	success
1 2 3 3 2 1	Yes	Testing the enqueue operator: front: 1 back: 1 front: 1 back: 2 front: 1 back: 3 front: 1 back: 3 front: 1 back: 2 front: 1 back: 1 Symmetric	success
1 2 3 4 3 2 5	No	Testing the enqueue operator: front: 1 back: 1 front: 1 back: 2 front: 1 back: 3 front: 1 back: 4 front: 1 back: 3 front: 1 back: 2 front: 1 back: 5 Not symmetric	success

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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